

CODE SPORT



In this month's column, we discuss the answers to some of the questions from last month.

In last month's column, we had discussed a few interview questions. While I received solutions from a few readers, I also received a couple of requests to discuss some of the concepts related to the questions. So let's do that. Instead of giving canned answers to the questions, I will provide the outline of the solution and ask readers to work things out.

Our first question from last month's column was on the choice of deployment architecture – whether one should select a scale-up vs scale-out solution. The first thing to remember is that the availability of hardware platform, the total cost of the solution and the software architecture of the solution are all inter-dependent. Any specific constraints or preferences in one of these three factors will impact the choice of the other two. For instance, if there is a constraint that the hardware platform available is more suitable as a scale-up platform, the software solution needs to be optimal for a scale-up system. This would typically mean a multi-threaded solution in which the data is shared implicitly through shared memory, instead of a 'shared nothing' scale-out software architecture where explicit data communication across the compute nodes would be required.

The question stated that the first choice of hardware was an enterprise class 16 core server with a huge 2TB RAM. Given that this is a system which is typically designed for scale-up solutions, as it has a large amount of physical memory that can be shared across the different application instances, the first preference would be to design a multi-threaded shared memory communication based solution for intrusion detection. The alternative hardware choice was multiple standard workstations with 32GB RAM. This hardware set-up is more amenable to a 'scale out' shared nothing solution, since each individual workshop is thin in computing resources and hence running a scale-up solution would be difficult. On the other hand, given the large array of workstations connected in a cluster, with each workstation having a small physical memory, it makes

more sense to host individual application instances on the cluster nodes, which can communicate through the network, instead of shared memory.

While this differentiation may suffice as an initial explanation, it does not cover all the issues raised in the question. It was mentioned that the proposed solution needs to monitor close to 256 systems and analyse their network logs for intrusion detection. It is not clear whether this analysis needs to be done in real-time or whether it is a retrospective analysis. The interviewer would definitely give additional brownie points if the candidate pointed out the different requirements that would arise if the analysis has to be done in real-time.

Another important perspective is to consider the possibility that a 'scale-up' and a 'scale-out' solution need not be mutually exclusive. For instance, it is possible to have a hybrid solution whereby individual nodes have a 'scale-up' solution running, with multiple instances being hosted in a 'scale-out' manner across the nodes. For instance, it is possible to have a 'scale-up' multi-threaded solution written using openMP API (www.openmp.org) and multiple instances of such openMP solutions are then tied together in a scale-out manner using Message Passing Interface (MPI). The ability to identify when such hybrid models would be efficient, and pointing out the trade-offs involved between a pure scale-up/scale-out and a hybrid compute model would definitely be a plus point in the eyes of the interviewer.

The second question in last month's column was actually about feature selection, though the term was never mentioned in the question. Given the large number of attributes/features present in the training data and given the constraint of a smaller physical memory system, one of the first answers the interviewer would be looking for is feature selection, wherein the number of features can be pruned to a smaller set of relevant features, which reduces the computational complexity of training the machine learning model on the given hardware. Since the training data contains 10,000